Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/05/2014
PI Name:	Lawler, John Ph.D.		
Project Title:	Redox Regulation of nNOS Translocation and Muscle Atrophy During Mechanical Unloading		
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Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843	**Congressional District:**	17
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/23/2016 per A. Chu/ARC and NSSC information (Ed., 6/16/15) NOTE: Grant number changed to NNX13AE45G, per A. Chu via NSSC (Ed., 1/29/13) NOTE: Grant number changed to NNX12AR62G, per A. Chu via NSSC (Ed., 9/28/12) NOTE: Period of performance (previously 6/27/12-6/26/15) and grant number (previously NNX12AL23G) changed per A. Chu/ARC and NSSC information (Ed., 9/12/2012)		
Key Personnel Changes/Previous PI:	No key personnel changes		
COI Name (Institution):	Fluckey, James Ph.D. (Texas A&M University)		
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Task Description:	A critical limitation to astronaut performance during spaceflight missions including those to the International Space Station and Mars is the impaired contractile ability of skeletal muscles. Skeletal muscle is a highly specialized tissue that rapidly adapts to dynamic changes in mechanical loading by modifying its ultrastructure and mass. Detection and transmittance of loading are vital to muscle integrity and protein turnover, with a nexus associated with the subsarcolemmal cytoskeleton that includes the dystrophin-glycoprotein complex (DGC) and caveolae. This interface between myocytes and the extracellular matrix confers mechanical integrity and activates cell signaling pathways that regulate protein turnover, mitochondrial, muscle fiber cross-sectional area, and fiber phenotype. Mechanical unloading experienced in the microgravity of spaceflight elicits large decrements in force generating capacity via reduction in muscle fiber cross sectional area or atrophy. While unloading-induced atrophy is a function of reduced protein synthesis coupled with increased protein degradation, activation of the transcription factors FoxO3a and nuclear factor-kappaB have drawn significant attention as mediators. Although the upstream triggers are not fully understood, translocation of neuronal nitric oxide synthase (nNOS) from the DGC and sarcolemma to the sarcoplasm is a novel signaling event that stimulates proteolysis, and thus atrophy. New clues can be gleaned from Duchenne and many of the limb-girdle muscular dystrophies, pathologies involving mutations to DGC, and other sarcolemmal proteins. Because MDs and microgravity both increases susceptibility of muscles to damage, oxidative stress, and pro-inflammatory signaling, a shared root cause is suspected: disruption of the subsarcolemmal cytoskeleton. New studies and Preliminary Data indicate the importance of NAD(P)H oxidase (NOX), a sarcolemmal source of oxidative stress, and caveolin-3 in stimulating myopathy in muscular dystrophy models. NOX may be activated upstream by angiotensin II and c-abl, a tyrosine kinase. Further, our pilot data show that unloading-induced alterations in caveolin-3 and nNOS are redox dependent and linked to atrophy and shift in muscle fiber-type. Our central hypothesis is that oxidative stress produced by NOX pathways and mitochondria directly contribute to a sarcolemmal loss of nNOS during mechanical unloading, enhancing proteolysis, and suppressing protein synthesis. The rodent hindlimb unloading model and novel fractional synthesis and degradation will be used to test our hypotheses. Specific Aim 1 will Identify NOX-dependent signaling on nNOS translocation and muscle fiber atrophy with hindlimb unloading. Specific Aim 2 will determine directly the role of mitochondrial and cytosolic oxidative stress in movement of nNOS away from the muscle cell membrane and disruption of DGC with mechanical unloading. In short, our primary objective is to identify the trigger by which skeletal muscles switch from being an anabolic to a catabolic machine during spaceflight, causing muscle atrophy and weakness. We propose that oxidative stress produced by a NOX pathway and mitochondria move the master switch protein (nNOS) away from the cell membrane, which slows protein synthesis and accelerates breakdown of protein. Our study would directly address high-priority gaps in Recapturing a Future for Space Exploration including the role of reactive oxygen species in protein balance and maintaining the slow contractile phenotype. Our grant proposal would also forward our understanding of (1) alterations in the subsarcolemmal cytoskeleton and membrane domains, (2) sensing and transduction of loading and gravity, (3) protein turnover, and (4) regulatory mechanisms that govern alterations in skeletal muscle, all programmatic emphases and goals of the current Space Biology NRA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bedrest, casting, and spaceflight cause significant skeletal muscle atrophy and weakness as a result of disuse. Prolonged spaceflight presents great challenges to the physical performance and health of astronauts as a result of microgravity, or reduced gravitational forces (i.e., loads). Mechanical unloading and extreme disuse during spaceflight produce rapid and dramatic skeletal muscle and weakness. Prolonged disuse also predisposes skeletal muscle to damage when landing on Mars, in an EVA (extra-vehicular activity), or returning to Earth. Furthermore, chronic disuse also contributes to and exacerbates underlying chronic pathology, including insulin resistance and cardiovascular disease. As a result crew health, safety, and completion of mission objectives are placed in jeopardy. While the pathways that cause atrophy and weakness due to disuse remain uncertain, proteins that sense changes in forces that skeletal muscles must produce may be the key. One of these proteins is nitric oxide synthase (nNOS), which is positioned near the muscle cell membrane. During spaceflight as well as aging and some types of muscular dystrophy, nNOS appears to be transported away from the cell membrane where it transforms from a muscle building to a muscle wasting agent. We will focus on oxidative stress, produced by excess free radicals produced by proteins near the cell membrane (ex. NOX) and mitochondria and how they affect nNOS transport as well as skeletal muscle atrophy. Understanding how these key membrane proteins are regulated will not only aid us in our treatment of disuse-related atrophy, but also will allow development of new therapeutics combating sedentary lifestyle and possible application to aging, Type 2 diabetes, and muscular dystrophies. Indeed, changes that are altered during spaceflight and ground models of spaceflight also appear to change with aging and diseases that cause skeletal muscle wasting and weakness.
	We have made significant progress on both Specific Aims 1 & 2 over the pass 2 years. This includes completion of the following experiments: (1) "Administration of the superoxide dismutase/catalase mimetic, Eukarion-134, prevents short-duration (54hr) unloading induced atrophy." We have recently published a paper in the American Journal of Physiology: Regulatory, Integrative, and Comparative Physiology entitled, “EUK-134 ameliorates nNOSμ translocation and skeletal muscle fiber atrophy during short-term mechanical unloading.” In this paper, we demonstrate that administration of the cell permeable synthetic superoxide dismutase (SOD)/catalase mimetic, EUK-134, prevents a reduction in Type I fiber cross sectional area following a 2-day (~54 hour) unloading protocol in the absence of changes in soleus muscle mass. In conjunction with a reduction of Type I fiber atrophy, EUK-134 prevented a shift in fiber type from type I to type II, an effect commonly observed with mechanical unloading. In addition, EUK-134 was associated with reduced expression of NOX-2 subunits, ROS production (dihydroethidium assay), and reduced ROS-related signaling/damage (4-hydroxynonenal). The positive effects of EUK-134 may be also related to a reduction in nNOS translocation from the sarcolemma to the cytosol and inactivation of the proteolytic signaling molecule, Fox-O3a. (2) "Administration of EUK-134 prevents 7-day unloading induced atrophy." We continued experimentation with EUK-134 to include a longer duration 7-day unloading protocol and expect to finalize this project soon. The findings from this study indicate that EUK-134 is also effective at ameliorating soleus muscle atrophy following 7-days of hindlimb unloading. Consistent with improved soleus muscle mass, fiber cross-sectional area was improved relative to the unloaded animals with EUK-134 administration; although, it did remain reduced relative to controls. Similarly to the animals that underwent the 2-day unloading protocol, animals with 7-day EUK-134 treatment had reduced markers of ROS production (3-nitrotyrosine) and inhibited nNOS translocation. For this subset of 7- day EUK-134 treated animals,

Task Progress:	we also measured expression of proteins involved in protein synthesis including protein kinase b (Akt) and mTOR. Activated (phosphorylated) Akt relative to total Akt was not different from controls with EUK-134 treatment, whereas activated (phosphorylated) mTOR relative to total mTOR and remained unchanged between all groups. (3) "Novel, targeted countermeasures to prevent ROS-induced skeletal muscle atrophy with unloading." In addition to study conducted with EUK-134, we have started and hope to complete studies in the near future involving more specific targeting of ROS production. This includes utilizing a NOX-2 inhibitor, gp91 ds-tat, and mitochondrial ROS inhibitor, SS-31. In doing so, we are trying to demonstrate which is the more effective inhibitor of ROS production with unloading as related to skeletal muscle alterations with unloading, determine the predominate source of ROS production, as well as evaluate if an interaction exists between these two sources of ROS. So far, utilizing the SS-31 mitochondrial ROS inhibitor has shown that it limits soleus muscle mass wasting and improves cross sectional area. Interestingly, SS-31 also decreases expression of the NOX-2 subunit, p67phox, suggesting a potential interaction between the two. As far as NOX-inhibition with gp91 ds-tat, we have also observed a decrease in muscle wasting as well as an improvement in muscle fiber cross sectional area. This is in conjunction with a reduction in ROS production (dihydroethidium assay). Future study will focus on combined therapy of gp91 ds-tat and SS-31 as well as delineating pathways responsible for limited atrophy with both treatments independently and combined. The effect of gp91 ds-tat and SS-31 on nNOS translocation will also be determined. (4) "Translating findings to spaceflight using nutraceutical and pharmaceutical interventions." Nutraceutical and pharmaceutical interventional studies have been commenced to mimic the effects of the ROS peptide inhibitors while developing more translatable findings. More specifically, we have begun to utilize the angiotensin II receptor blocker, losartan, commonly used clinically, which should prevent upstream activation of NOX- 2 by inhibiting angiotensin II signaling. Our preliminary findings suggest that soleus muscle mass is not affected by losartan treatment during unloading; however, fiber cross sectional area was no longer different from control with unloading. In addition, losartan prevented fiber atrophy in type I fibers and a shift in fiber type with unloading. Losartan treatment is also apparently effective in decreasing NOX-2 expression and preventing translocation of nNOS from the sarcolemma to the cytosol. A nutraceutical intervention with a combined treatment of fish oil and curcumin is also apparently effective in preventing soleus muscle atrophy and improving cross sectional area. Future investigation will further clarify the effectiveness of losartan in treating unloading-induced atrophy as well as the fish oil + circumin intervention. (5) "Overall summary of future studies." Future study will focus on finalizing the 7-day EUK-134 interventional study including measurement of fractional synthesis rate; further delineating the mechanisms responsible for reduced atrophy with both gp91ds-tat and SS-31 interventions; utilizing a genetic ablation of NOX-2 (gp91phox-/-) and/or dysferlin to further establish their role in unloading induced atrophy; potentially determining the role of microtubule disruption in unloading induced atrophy and nNOS translocation; and establishing the effectiveness of pharmaceutical and nutraceutical interventions in preventing unloading-induced atrophy.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Lawler JM, Lee Y, Hord JM, Guzzoni V, Kuczmarski JM. "Mitochondrial ROS Amplify Nox2 Signaling and Atrophy in the Rat Soleus with 7 Days of Hindlimb Unloading." Presented at Experimental Biology 2014, San Diego, CA, April 26-30, 2014. FASEB Journal. 2014 Apr;28(1 Suppl):LB821. See also http://www.fasebj.org/content/28/1_Supplement.toc for searching. , Apr-2014
Abstracts for Journals and Proceedings	Hord JM, Lee Y, Lawler JM. "Angiotensin II receptor blockade protects against the slow to fast fiber type shift and type I fiber atrophy in the rat soleus with 7 days of hindlimb unloading." Presented at Experimental Biology 2014, San Diego, CA, April 26-30, 2014. FASEB Journal. 2014 Apr;28(1 Suppl):LB804. See also http://www.fasebj.org/content/28/1_Supplement.toc for searching. , Apr-2014
Abstracts for Journals and Proceedings	Lee Y, Lawler MS, Botchlett R, Hord JM, Lawler JM. "Dysregulation of dysferlin during 7 days mechanical unloading is mediated by ROS in rat soleus muscle." Presented at Experimental Biology 2014, San Diego, CA, April 26-30, 2014. FASEB Journal. 2014 Apr;28(1 Suppl):1102.27. See also http://www.fasebj.org/content/28/1_Supplement.toc for searching. , Apr-2014
Abstracts for Journals and Proceedings	Hord JM. "Inactivity-induced skeletal muscle atrophy: role of 12/15-Lipoxygenase." Presented at the Texas Chapter of the American College of Sports Medicine Meeting, Fort Worth, TX, February 27-28, 2014. Texas Chapter of the American College of Sports Medicine Meeting, Fort Worth, TX, February 27-28, 2014. , Feb-2014
Abstracts for Journals and Proceedings	Lee Y. "Role of Nox2 in disuse-induced skeletal muscle atrophy." Presented at the Texas Chapter of the American College of Sports Medicine Meeting, Fort Worth, TX, February 27-28, 2014. Texas Chapter of the American College of Sports Medicine Meeting, Fort Worth, TX, February 27-28, 2014. , Feb-2014
Articles in Peer-reviewed Journals	Wesolowski LT, Simons JL, Semanchik PL, Othman MA, Kim J-H, Lawler JM, Kamal KY, White-Springer SH. "The impact of SRT2104 on skeletal muscle mitochondrial function, redox biology, and loss of muscle mass in hindlimb unloaded rats." Int J Mol Sci. 2023 Jul 6;24(13):11135. https://doi.org/10.3390/ijms241311135 ; PMID: 37446313; PMCID: PMC10342025 , Jul-2023
Articles in Peer-reviewed Journals	Kamal KY, Othman MA, Kim J-H, Lawler JM. "Bioreactor development for skeletal muscle hypertrophy and atrophy by manipulating uniaxial cyclic strain: Proof of concept." npj Microgravity. 2024 Jun 11;10:62. https://doi.org/10.1038/s41526-023-00320-0 ; PMID: 38862543; PMCID: PMC11167039 , Jun-2024

Articles in Peer-reviewed Journals	Mason CE, Green J, Adamopoulos KI, Afshin EE, Baechle JJ, Basner M, Bailey SM, Bielski L, Borg J, Borg J, Broddrick JT, Burke M, Caicedo A, Castañeda V, Chatterjee S, Chin C, Church G, Costes SV, De Vlaminck I, Desai RI, Dhir R, Diaz JE, Etlin SM, Feinstein Z, Furman D, Garcia-Medina JS, Garrett-Bakelman F, Giacomello S, Gupta A, Hassanin A, Houerbi N, Irby I, Javorsky E, Jirak P, Jones CW, Kamal KY, Kangas BD, Karouia F, Kim J, Kim JH, Kleinman A, Lam T, Lawler JM, Lee JA, Limoli CL, Lucaci A, MacKay M, McDonald JT, Melnick AM, Meydan C, Mieczkowski J, Muratani M, Najjar D, Othman MA, Overbey EG, Paar V, Park J, Paul AM, Perdyan A, Proszynski J, Reynolds RJ, Ronca AE, Rubins K, Ryon KA, Sanders LM, Glowe PS, Shevde Y, Schmidt MA, Scott RT, Shirah B, Sienkiewicz K, Sierra M, Siew K, Theriot CA, Tierney BT, Venkateswaran K, Hirschberg JW, Walsh SB, Walter C, Winer DA, Yu M, Zea L, Mateus J, Beheshti A. "A second space age spanning omics, platforms, and medicine across orbits." Nature. 2024 Aug;632(8027):995-1008. Review. https://doi.org/10.1038/s41586-024-07586-8 ; PubMed PMID: 38862027; PubMed Central PMCID: PMC12366838 , Aug-2024
Articles in Peer-reviewed Journals	Lawler JM, Kamal KY, Botchlett RE, Woo SL, Li H, Hord JM, Fluckey JD, Wu C. "Metformin ablates high fat diet-induced skeletal muscle hypertrophy and elevation of sarcolemmal GLUT4 when feeding is initiated in young adult male mice." Connect Tissue Res. 2025 Mar 7;1-15. https://doi.org/10.1080/03008207.2025.2471853 ; PMID: 40052722 , Mar-2025
Articles in Peer-reviewed Journals	Lawler J, Kim J, Othman M, Kamal K, Fluckey J. "The sirtuin-1 agonist SRT2104 mitigates redox signaling and nNOS translocation in unloaded skeletal muscle." Physiology. 2025 May 12;40(S1):1343. https://journals.physiology.org/doi/abs/10.1152/physiol.2025.40.S1.1343 , May-2025
Articles in Peer-reviewed Journals	Othman M, Kim J, Kamal K, Lawler J. "RANKL elevation contributes to skeletal muscle myopathy in aging: Unveiling of novel mechanisms." Physiology. 2025 May 12;40(S1):0567. https://journals.physiology.org/doi/abs/10.1152/physiol.2025.40.S1.0567 , May-2025
Articles in Peer-reviewed Journals	Lawler JM, Kunst M, Hord JM, Lee Y, Joshi K, Botchlett RE, Ramirez A, Martinez DA. "EUK-134 ameliorates nNOSμ translocation and skeletal muscle fiber atrophy during short-term mechanical unloading." Am J Physiol Regul Integr Comp Physiol. 2014 Apr 1;306(7):R470-82. Epub 2014 Jan 29. http://dx.doi.org/10.1152/ajpregu.00371.2013 ; PubMed PMID: 24477538 ; PubMed Central PMCID: PMC3962621 , Apr-2014
Articles in Peer-reviewed Journals	Lawler JM, Kwak HB, Kim JH, Lee Y, Hord JM, Martinez DA. "Biphasic stress response in the soleus during reloading after hind limb unloading." Medicine and Science in Sports & Exercise. 2012 Apr;44(4):600-9. http://dx.doi.org/10.1249/MSS.0b013e31823ab37a ; PubMed PMID: 21983076 , Apr-2012
Awards	Lee Y. "2014 ACSM Doctoral Student Research Grant. Nox2 Regulation in Muscle Atrophy during Disuse. American College of Sports Medicine, January 2014." Jan-2014
Awards	Kuczmarski JM. "Postdoctoral Fellowship. Developing Novel, Targeted Countermeasures to Reduce Oxidative Stress and Skeletal Muscle Atrophy During Microgravity. National Space and Biomedical Research Institute, December 2013." Dec-2013
Awards	Hord J. "2014-2015 College of Education and Human Development Graduate Research Fellowship. Texas A&M University, January 2014." Jan-2014
Awards	Hord J. "2014 Doctoral Student Research Development Award. Texas Chapter of the American College of Sports Medicine, January 2014." Jan-2014
Awards	Lee Y. "2014 Doctoral Student Research Development Award. Texas Chapter of the American College of Sports Medicine, January 2014." Jan-2014
Awards	Lee Y. "2013-2014 Huffines Institutional Student Grant. Role of Nox2 in Diaphragm Pathology with Duchenne Muscular Dystrophy, January 2013." Jan-2013
Awards	Botchlett R. "2012-2013 Huffines' Institute Student Grant. The effects of a combination treatment of fish oil and curcumin on reducing unloading-induced atrophy, January 2012." Jan-2012
Books/Book Chapters	Botchlett R, Lawler JM, Wu G. "L-Arginine and citrulline in sports nutrition and health." in "Nutrition and Enhanced Sports Performance: Muscle Building, Endurance, and Strength." Ed. D. Bagchi, S. Nair, C.K. Sen. Amsterdam ; Boston : Elsevier/AP, Academic Press Academic Press, 2013. p. 439-446. http://dx.doi.org/10.1016/B978-0-12-396454-0.00045-X , Aug-2013